BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings illustrate various embodiments of the systems and methods described herein and are not intended to limit the scope of the disclosure.

FIG. 1 illustrates a perspective view of an example handheld penile geometry scanning device comprising a housing structure, at least one time-of-flight (ToF) ranging sensor, and at least one optical motion sensing component positioned to detect longitudinal displacement relative to the anatomical surface.

FIG. 2 illustrates a cross-sectional view of an embodiment in which a ToF sensor is positioned within a reference frame configured to measure radial distance between the sensor plane and the penile surface during controlled scanning motion.

FIG. 3 illustrates an embodiment of a C-shaped or partially enclosed scanning structure configured to maintain a defined spatial orientation relative to the anatomy while capturing sequential distance samples along the longitudinal axis.

FIG. 4 illustrates an embodiment incorporating an optical flow sensor or speckle-based motion detection module configured to track displacement of the device relative to the anatomical surface.

FIG. 5 illustrates an embodiment further comprising an inertial measurement unit (IMU) configured to detect tilt, angular deviation, or unintended motion and to provide correction data to a reconstruction processor.

FIG. 6 illustrates a schematic block diagram of a processing architecture comprising distance acquisition circuitry, motion tracking modules, a computational reconstruction engine, and a geometry profile generation module.

FIG. 7 illustrates an example reconstructed three-dimensional geometric model derived from sequential ToF distance samples and motion-tracked positional data.

FIG. 8 illustrates representative derived structural metrics generated from the reconstructed geometry profile, including longitudinal length, cross-sectional radius distribution, curvature mapping, taper gradients, volumetric estimation, and proportionality indices.

FIG. 9 illustrates an embodiment in which the generated geometry profile interfaces with a digital storage system, calibration module, compatibility modeling system, or broader biometric analytics platform.

It will be understood that the illustrated embodiments are exemplary and that alternative housing configurations, sensor arrangements, motion tracking implementations, reconstruction algorithms, and integration architectures may be utilized without departing from the spirit and scope of the present disclosure.